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Systems Engineer and Data Engineer. I build tools that let AI operate production infrastructure: a self-built fleet of MCP servers operated from Claude Code, and ETL pipelines with Dagster/Python over a SQL Server DWH with autonomous monitoring and multi-channel alerting. Cut a monthly forecast from ~18h to ~61 min via multiprocessing.

WORK EXPERIENCE

Grupo Farinter

RPA Developer & Data Engineer | November 2025 - Present

- Cut a monthly demand forecast (~23,000 series) from ~18 hours to ~61 minutes in production, parallelizing with ProcessPoolExecutor (dead-series pruning, byte-identical and deterministic output) under strict TDD with no new dependencies.
- Built and operate a self-built fleet of MCP (Model Context Protocol) servers —Dagster, SQL Server, SMB, Dokploy, Microsoft Graph/Outlook— with a read-only validator, token optimization (~10-20x) and a multi-agent adversarial audit, applying Agentic AI (sub-agents, Agent Skills, hooks) under SDD and strict TDD.
- Developed bank reconciliation pipelines (Credomatic/BAC, Ficohsa) and transactional pipelines (Netflix, Tengo, Rapibac) over SMB and SQL Server, with file-sensor detection and incremental loading.
- Built a full-stack Python (Reflex) competitive pricing dashboard comparing Kielsa's margins against ~7,700 competitor SKUs across 8 analytical views, with SCD2 margin-erosion detection and suggested-price simulation; 801 tests and CI/CD to Dokploy.
- Built an autonomous watchdog that detects and re-launches stuck pipelines (baselines from 100+ runs, Telegram notifications, timeouts on 27+ jobs) and redesigned failure alerts to Discord/email with strict TDD and graceful degradation.
- Closed a data-quality gap with a generic AssetCheckEvaluation sensor that alerts only on errors to Discord/Telegram/email (cursor-based dedup); it surfaced a silent 182,928-row duplicate that was passing with all runs green.
- Built competitor price scrapers with daily extraction in Dagster, reverse engineering of obfuscated APIs and proxy/TOR rotation.
- Built hvault, a pure-Rust service (CLI + daemon) syncing Traefik TLS certificates across nodes via HashiCorp Vault (KV-v2), with atomic writes for zero-downtime hot-reload, x509 validation and a Leptos/WASM admin UI with OIDC.
- Automated cash-closing flows, migrated legacy Windows processes to Dagster and standardized DWH tables with metadata (country, source, versions) for Power BI consumption.

Tecnologías: Python, Dagster, SQL Server, DWH, Docker, SMB, Discord Webhooks, dbt, MCP, Model Context Protocol, Claude Code, Codex CLI, OpenCode, Cursor, Aider, Cline, Continue.dev, Windsurf, Gemini CLI, GitHub Copilot, Agent Skills, SKILL.md, Slash Commands, Hooks, AI Agents, Agentic AI, Sub-agents, Agent Orchestration, Tool Use, Function Calling, AI Integration, LLMs, GenAI, Anthropic API, OpenAI API, SDD, Spec-Driven Development, TDD, Prompt Engineering, Git, statsforecast, multiprocessing, ProcessPoolExecutor, concurrent.futures, smbprotocol, Dokploy, Data Quality, SCD2, Reflex, Power BI, Rust, HashiCorp Vault

Analiza Clinical Laboratories

Software Developer | March 2025 - October 2025

- Automated VITEK-CRM integration with FastAPI and ASTM/HL7 protocols in production for bidirectional processing of culture results.
- Modeled catalogs with SQLAlchemy ORM (7,429 microorganisms, 278 antibiotics) with complex relationships between antibiogram headers and detail records.
- Built a corporate proxy with Docker, Squid and OpenVPN, reducing management of 30+ machines to a single point.

Tecnologías: Python, FastAPI, TypeScript, NestJS, PostgreSQL, SQLAlchemy, TypeORM, Docker, WebSocket, PowerShell, ASTM, HL7, VITEK

GuabaBIT

QA & Backend Developer | October 2024 - March 2025

- Designed the architecture of a community payments system (NestJS, DynamoDB, AWS S3/SNS) and implemented E2E tests with Playwright for critical flows.

SKILLS

- Languages: Python, TypeScript, JavaScript, C++, Rust, SQL, T-SQL
- AI & AI-Assisted Development: Claude Code, Cursor, GitHub Copilot, MCP (Model Context Protocol), Anthropic Agent Skills, AI Agents, Sub-agents, Tool Use / Function Calling, Prompt Engineering, SDD (Spec-Driven Development), Anthropic API, LLMs / GenAI
- Frameworks & Orchestration: Dagster, FastAPI, NestJS, Reflex, Astro
- Databases: SQL Server, PostgreSQL, MongoDB, DynamoDB, SQLite
- Cloud & Infrastructure: AWS (S3, SNS, DynamoDB), Azure, Docker, Terraform, Dokploy, HashiCorp Vault
- Testing & QA: Playwright, Jest, Postman, Apidog, Jira
- Data Engineering: dbt, Power BI, ETL, Web Scraping, DWH, Polars, statsforecast, Data Quality

PROJECTS

MCP Servers Suite — Self-built production fleet

Fleet of production MCP (Model Context Protocol) servers operated from Claude Code with Tool Use / Function Calling: mcp-dagster (assets/runs via GraphQL), mcp-db (Jinja-templated queries + read-only validator over SQL Server/PostgreSQL), mcp-api-check (health checks), smb-mcp (SMB file ops), mcp-dokploy (deployments) and outlook-todoist-mcp (Outlook → Todoist via Microsoft Graph). Modernized with token optimization (~10-20x), a multi-agent adversarial audit and monolith splitting.

Tecnologías: Python, MCP, Dagster GraphQL, SQL Server, PostgreSQL, Microsoft Graph, smbprotocol, Dokploy, Claude Code

Dagster Watchdog — Autonomous Pipeline Monitoring

Autonomous system that polls Dagster via GraphQL every 5 minutes, detects stuck runs by comparing against learned baselines, cancels them, re-launches with retry tags and notifies via Telegram. Includes safety guards and runs as a 24/7 systemd service.

Tecnologías: Python, Dagster GraphQL, Telegram Bot, systemd, asyncio, Observability

Accounting System — Azure Infrastructure with Terraform

Cloud infrastructure on Azure as Infrastructure as Code: Virtual Network, SQL Database, Storage and App Service with advanced security measures (private endpoints, NSGs, managed identities).

Tecnologías: Azure, Terraform, IaC, Azure SQL, Azure App Service, Cloud Architecture

Automated YouTube Channel — Multimedia Pipeline

End-to-end pipeline: royalty-free music scraping, editing with VHS effects, GPU-accelerated 4K rendering (CUDA), playlist management and scheduled uploads via the YouTube API with copyright handling.

Tecnologías: Python, C++, CUDA, FFmpeg, Moviepy, Playwright, YouTube API, Web Scraping

EDUCATION

National Autonomous University of Honduras

B.S. in Mathematics (in progress) | Tegucigalpa, Honduras | 2024 - Present

National Autonomous University of Honduras

B.S. in Computer Systems Engineering | Tegucigalpa, Honduras | 2019 - 2025

CERTIFICATIONS

- Oracle Next Education F2 T5 Back-end - Alura Latam, 2023
- Database Foundations - Oracle Academy, 2023

LANGUAGES

- Spanish: Native
- English: B2 Level